

MEDS Lab Module 4

Section 5: Getting Started on EDA Playground

Task 1: Hello Gate

July 23, 2026

1. EDA Playground Link

The complete design and testbench code for the AND, OR, and XOR gate implementations can be found at the following shareable link:

- <https://edaplayground.com/x/t8gC>

2. Schematic Diagram

Below is the gate-level schematic diagram representing the multi-gate implementation.

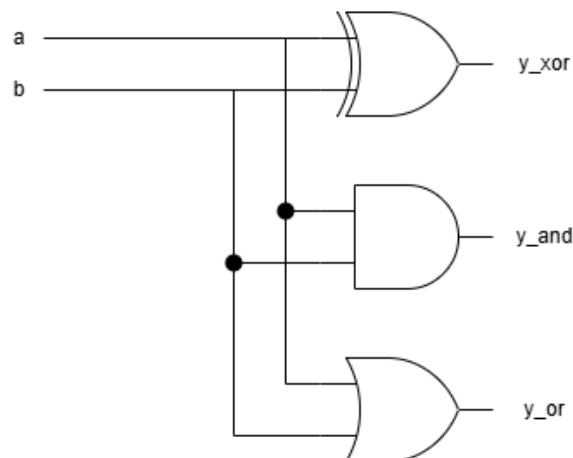


Figure 1: Gate-level schematic diagram for the AND, OR, and XOR functions.

3. Terminal Output Screenshot

The following screenshot displays the simulation log and the `$display` output, confirming the expected values for the tested input combinations.

```
Compiler version X-2025.06-SP1_Fu1164; Runtime version X-2025.06-SP1_Fu1164;  Jul 20 01:53 2026
AND Gate Output
a=0 b=0 y=0
a=0 b=0 y=0
a=0 b=0 y=0
AND Gate Output
a=1 b=0 y=0
a=1 b=0 y=1
a=1 b=0 y=1
AND Gate Output
a=1 b=1 y=1
a=1 b=1 y=1
a=1 b=1 y=0
VCS Simulation Report
Time: 30 ns
CPU Time:      0.430 seconds;      Data structure size:  0.0Mb
Mon Jul 20 01:53:34 2026
Done
```

Figure 2: Simulation log output from EDA Playground.

4. Waveform Screenshot

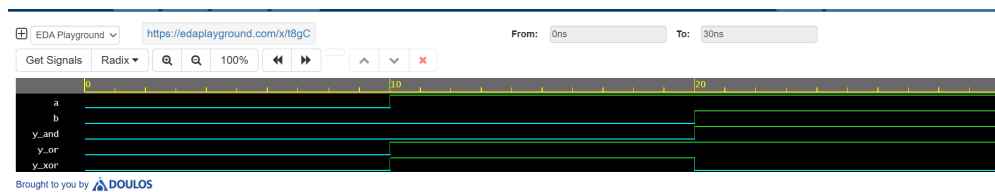


Figure 3: EPWave waveform confirming output behavior.